Partial Derivatives and the Gradient

Section 1: First-Order Partial Derivatives

1 Find f_x and f_y for $f(x, y) = 3x^2y - 5xy^3 + 4x - 2y$.

a. $f_x = 6xy - 5y^3 + 4$, $f_y = 3x^2 - 15xy^2 - 2$

b. $f_x = 6xy - 5y^3$, $f_y = 3x^2 - 15xy^2$

c. $f_x = 6x - 5y^3 + 4$, $f_y = 3x^2 - 15y^2 - 2$

d. $f_x = 3x^2 - 15xy^2$, $f_y = 6xy - 5y^3$

2 Find $\frac{\partial z}{\partial x}$ for $z = e^{x^2y}$.

a. $2xye^{x^2y}$

b. $x^2e^{x^2y}$

c. $2xe^{x^2y}$

d. $2xye^{2xy}$

3 Find $f_y(1, 0)$ for $f(x, y) = \ln(x^2 + y^2 + 1)$.

a. 0

b. 1

c. 2

d. 1/2

Section 2: Second-Order Partials and Clairaut's Theorem

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4 What does Clairaut's Theorem (equality of mixed partials) state for continuous second partial derivatives?

a. $f_{xy} = f_{yx}$

b. $f_{xx} = f_{yy}$

c. $f_{xy} = -f_{yx}$

d. $f_{xx} + f_{yy} = 0$

5 Find f_{xx} for $f(x, y) = x^4y^2 - 2x^2y$.

a. $12x^2y^2 - 4y$

b. $4x^3y^2 - 4xy$

c. $2x^4 - 0$

d. $12x^2 - 4$

6 Verify whether $u(x, y) = e^x \sin y$ satisfies Laplace's equation $u_{xx} + u_{yy} = 0$.

a. Yes, $u_{xx} + u_{yy} = e^x \sin y - e^x \sin y = 0$

b. No, $u_{xx} + u_{yy} = 2e^x \sin y$

c. No, it equals $e^x \cos y$

d. Only when $x = 0$

Section 3: The Gradient Vector and Directional Derivatives

7 Find the gradient $\nabla f(x, y)$ for $f(x, y) = x^2y + 3y^2$ at the point $(2, 1)$.

a. $\langle 4, 10 \rangle$

b. $\langle 4, 7 \rangle$

c. $\langle 2, 6 \rangle$

d. $\langle 10, 4 \rangle$

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8 In which direction does a differentiable function f increase most rapidly at a point P?

a. In the direction of the gradient vector $\nabla f(P)$

b. Opposite to $\nabla f(P)$

c. Orthogonal to $\nabla f(P)$

d. Along the level curve

9 Find the directional derivative of $f(x, y) = 2x^2 - 3y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.

a. $-12/5$

b. $12/5$

c. -24

d. 0

Section 4: Tangent Planes and Linear Approximations

10 Find the equation of the tangent plane to $z = x^2 + y^2$ at the point $(1, 2, 5)$.

a. $z - 5 = 2(x - 1) + 4(y - 2)$

b. $z - 5 = (x - 1) + 2(y - 2)$

c. $2x + 4y - z = 0$

d. $z = 2x + 4y + 5$

11 Use the linear approximation $L(x, y)$ of $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$ to estimate $f(3.02, 3.97)$.

a. 4.988

b. 5.012

c. 5.000

d. 4.950

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12 Find the normal vector to the level surface $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$.

a. $\langle 2, 4, 4 \rangle$

b. $\langle 1, 2, 2 \rangle$

c. Both $\langle 1, 2, 2 \rangle$ and $\langle 2, 4, 4 \rangle$ are valid normal vectors

d. $\langle 2, 2, 1 \rangle$

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